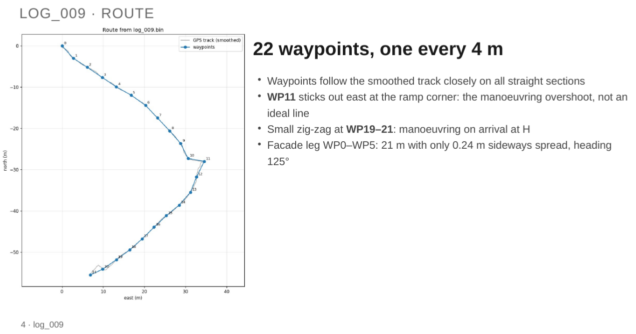
PRESENTER 4 of 8 · Route from log_009 · about 2.5 minutes



LOG_009 · ROUTE

22 waypoints, one every 4 m

- Waypoints follow the smoothed track closely on all straight sections
- **WP11** sticks out east at the ramp corner: the manoeuvring overshoot, not an ideal line
- Small zig-zag at **WP19–21**: manoeuvring on arrival at H
- Facade leg WP0–WP5: 21 m with only 0.24 m sideways spread, heading 125°

4 · log_009

GOAL: Explain how the 22 waypoints were computed and what their quality is.

WHAT TO SAY

From the log_009 drive we generated the route automatically with a Python tool called make_route.py. The plot shows the result: the grey line is the smoothed GPS track, the blue dots are the 22 waypoints, numbered 0 at Building B to 21 at Building H.

The method has four steps. First, we keep only real GPS updates. The GPS module updates once per second, but the log writes four records per second, so each GPS value appears about four times; we remove the repeats. Second, we smooth the positions with a moving average over five fixes, about five seconds, because single GPS fixes jump by a metre or more. Third, we convert latitude and longitude into metres east and north of the start point. Over a few hundred metres a simple flat-earth formula is accurate to centimetres: one degree of latitude is about 111 km, and one degree of longitude is that times the cosine of the latitude, here about 73 km. Fourth, we walk along the smoothed track and keep a point every time we are at least 4 metres from the last kept point.

Why 4 metres? GPS jitter at standstill was about plus or minus 1.5 metres, and the rover counts a waypoint as reached within 3 metres. The spacing must be larger than that radius, otherwise the next waypoint is already reached when the rover arrives at the current one.

How good is the route? On all straight sections the waypoints lie on the track. The first leg along the facade, WP0 to WP5, covers 21 metres with a sideways spread of only 0.24 metres and a direction of 125 degrees, so the rover drove very straight there. Two places need care. WP11 sticks out to the east at the ramp corner; that is where the rover manoeuvred, not where it should ideally drive, so WP11 is a candidate to delete. And the small zig-zag at WP19 to WP21 is manoeuvring on arrival at H.

Finally, the tool writes the waypoints directly as a C++ array, which we compile into the box firmware.

KEY FACTS TO REMEMBER

22 waypoints · every 4 m · 5-fix smoothing · facade leg 21 m, 0.24 m spread, 125° · WP11 = overshoot.

IF YOU ARE ASKED

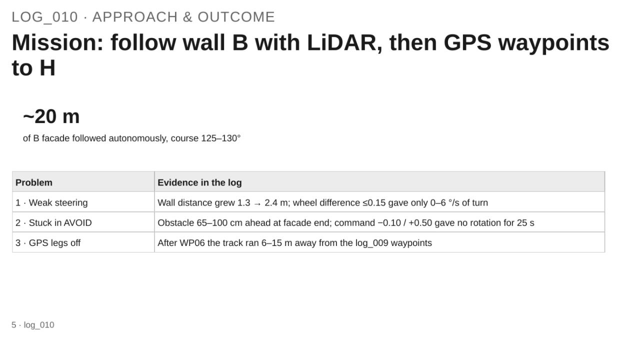
"Does smoothing cut corners?" Slightly; about 2 seconds of delay, which is why we check the plot by hand.

"How much memory does the route need?" 22 points of two numbers each, a few hundred bytes.

HANDOVER

"[Name] will now explain our first autonomous mission, log_010."

PRESENTER 5 of 8 · log_010 approach and outcome · about 3 minutes



LOG_010 · APPROACH & OUTCOME

Mission: follow wall B with LiDAR, then GPS waypoints to H

~20 m

of B facade followed autonomously, course 125–130°

Problem	Evidence in the log
1 · Weak steering	Wall distance grew 1.3 → 2.4 m; wheel difference ≤0.15 gave only 0–6 °/s of turn
2 · Stuck in AVOID	Obstacle 65–100 cm ahead at facade end; command −0.10 / +0.50 gave no rotation for 25 s
3 · GPS legs off	After WP06 the track ran 6–15 m away from the log_009 waypoints

5 · log_010

GOAL: Explain how the mission was designed, what worked, and the three problems.

WHAT TO SAY

log_010 was our first complete autonomous mission, built on the log_009 route.

The mission had two legs. Leg one: follow the facade of Building B with the LiDAR. The box fits a straight line through the LiDAR points on the left side. That line gives the distance to the wall and the angle of the rover relative to it. The controller steers to keep a set distance and to stay parallel. At the start the rover is placed parallel to the facade, and since we know the facade direction, 125 degrees, this also gives the rover its true heading. Leg two starts when the wall disappears: drive to the log_009 waypoints one after another. For each waypoint the box computes the direction to it from the GPS position, compares it with the rover heading, and steers by the difference.

Every command, in both legs, still passes the safety layer: no forward motion if an obstacle is closer than 25 cm by ultrasonic or 30 cm by LiDAR, and a stop if the tilt is too large.

What worked: the rover followed the B facade autonomously for about 20 metres, with a course of 125 to 130 degrees, exactly the facade direction.

Then three separate problems appeared, shown in the table.

First, weak steering. During the wall leg the distance to the wall grew from 1.3 to 2.4 metres. The controller asked for a correction, but the rover hardly turned. The reason is the skid-steered chassis: to turn, the tyres have to slide sideways, which needs a large difference between left and right wheel speeds. Our small corrections, at most 0.15, gave only 0 to 6 degrees per second.

Second, the rover got stuck. At the end of the facade something was 65 to 100 cm ahead. The avoidance manoeuvre sent minus 0.10 to one side and plus 0.50 to the other. That is neither a real turn on the spot nor a real arc, and the rover did not rotate for 25 seconds, so we took over by hand. Also, the switch to the GPS leg never happened, because the wall was still visible at 2.4 metres.

Third, the GPS legs were off: after waypoint 6, the track was 6 to 15 metres away from the waypoints. Slide 7 explains this in detail.

KEY FACTS TO REMEMBER

Leg 1 LiDAR wall following, heading seed 125° · Leg 2 GPS waypoints · about 20 m autonomous · three problems: steering, stuck in AVOID, GPS offset.

IF YOU ARE ASKED

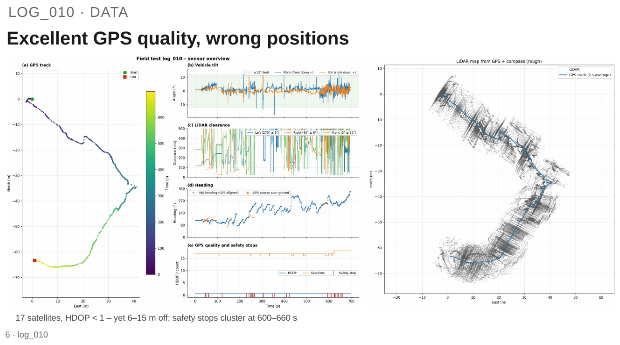
"Why not just increase the steering gain?" Below a certain wheel difference this chassis does not turn at all; the controller has to be designed around what the chassis can physically do.

"Was the Wi-Fi delay a problem?" No longer: median 50 ms, at most 120 ms in this run.

HANDOVER

"[Name] will now show the data behind these findings."

PRESENTER 6 of 8 · log_010 data · about 2.5 minutes



LOG_010 · DATA

Excellent GPS quality, wrong positions

17 satellites, HDOP < 1 – yet 6–15 m off; safety stops cluster at 600–660 s

6 · log_010

GOAL: Show the log_010 sensor data and explain why excellent GPS quality can still mean wrong positions.

WHAT TO SAY

These are the data of log_010, shown in the same five panels as log_009, so the two can be compared directly.

In panel (a), the GPS track, the run goes from B down to H, as before. In panel (b), the tilt stays mostly inside the 12-degree band. There are roll spikes of about 15 degrees near 240 and 300 seconds; those are kerbs or ramp edges. In panel (c), the LiDAR distances, the left wall distance grows during the first 100 seconds. That is the weak steering on the wall leg we just heard about: the rover slowly drifted away from the facade.

The most important panel is (e), GPS quality. The receiver saw 17 satellites, and HDOP stayed below 1 the whole time. By the usual numbers, that is excellent GPS. Yet, as the next slide will show, the positions were 6 to 15 metres away from the same path in log_009.

How can both be true? HDOP, horizontal dilution of precision, only describes the geometry of the satellites in the sky. Satellites spread across the sky give a low HDOP, satellites bunched together give a high one. It says nothing about errors in the signals themselves. Next to tall buildings, signals are reflected by the facades before they reach the antenna. That is called multipath. A reflected signal travels further, so the receiver computes a wrong distance to that satellite and a shifted position. Our NEO-6M receiver has no correction service such as RTK, so these errors stay in the position.

The red ticks in panel (e) are safety stops. They cluster at 600 to 660 seconds, near the end of the route, where the passage is narrow and the LiDAR and ultrasonics often saw obstacles close by.

On the right, the LiDAR map of log_010 shows the same wall shapes as in log_009. That is important for the comparison: the buildings did not move, so if the GPS tracks differ, the difference is in the GPS, not in the path.

KEY FACTS TO REMEMBER

17 satellites, HDOP < 1 · still 6–15 m off · HDOP = geometry only · multipath from facades · safety stops cluster at 600–660 s · same wall shapes as log_009.

IF YOU ARE ASKED

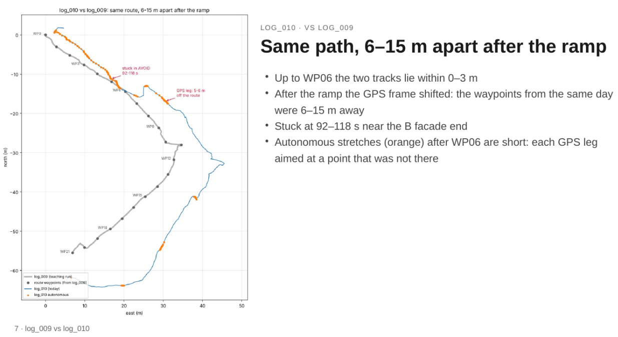
"What is HDOP exactly?" A factor by which satellite geometry multiplies the measurement error; below 1 is excellent, above 2.5 we would stop and wait.

"Could the receiver be broken?" Unlikely; within one run the track is smooth and consistent, the offset appears as a shift of the whole track.

HANDOVER

"[Name] will now compare the two runs directly."

PRESENTER 7 of 8 · log_010 compared with log_009 · about 3 minutes



LOG_010 · VS LOG_009

Same path, 6–15 m apart after the ramp

- Up to WP06 the two tracks lie within 0–3 m
- After the ramp the GPS frame shifted: the waypoints from the same day were 6–15 m away
- Stuck at 92–118 s near the B facade end
- Autonomous stretches (orange) after WP06 are short: each GPS leg aimed at a point that was not there

7 · log_009 vs log_010

GOAL: Present the main finding: the same physical path, 6–15 m apart in GPS, and how we measured it.

WHAT TO SAY

This plot is the main evidence of our talk. It shows both runs in one common coordinate frame.

The thick grey line is log_009, the manual reference run, with its waypoints as grey dots. The thin blue line is log_010. The orange dots mark the moments when the box drove on its own. The red notes mark two events: the rover stuck in the avoidance manoeuvre at 92 to 118 seconds near the end of the B facade, and a GPS leg aiming 5 to 6 metres off the route.

How was the plot made? Both logs are converted to metres around the same origin, waypoint 0 of log_009. This detail matters: if each run were centred on its own start point, a constant GPS offset between the runs would be hidden. To get a number, we took every fourth GPS fix of log_010 and measured its distance to the nearest point of the log_009 track, then summarised this per 50-second window. The orange autonomous points are identified from the wheel commands: the phone drive pad only sends fixed steps of 0.05 with both wheels equal or opposite; any other command came from the controller.

The result: up to waypoint 6, before the ramp, the two tracks lie within 0 to 3 metres of each other. After the ramp, they are 6 to 15 metres apart for the rest of the route, although the rover drove along the same physical path, which has only one way through: the raised path, the ramp, the gap between D and E.

We also tested whether this is a different path or a shifted one: the script searched for the best constant shift that moves log_010 onto log_009. After that shift the remaining distance was small. So the shape is the same and the whole track is displaced, which is what a GPS offset looks like, not random noise.

What does this mean for navigation? Waypoints recorded in one run cannot be trusted in another run. The GPS legs in log_010 steered towards points that, in that run's GPS frame, were somewhere else. This is why the autonomous stretches after waypoint 6 are short.

KEY FACTS TO REMEMBER

Common origin WP0 · nearest-point distance · 0–3 m before WP06, 6–15 m after · best shift leaves a small residual = offset, not noise.

IF YOU ARE ASKED

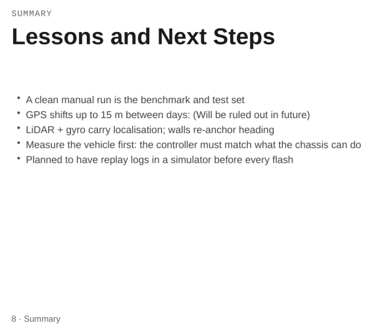
"Could you not just average more GPS fixes?" Averaging removes noise, but this is a bias that stays for minutes, so averaging does not remove it.

"Is one comparison enough?" It shows the effect can be this large; more runs would show how often it happens.

HANDOVER

"[Name] will close with our lessons and next steps."

PRESENTER 8 of 8 · Lessons and next steps · about 2.5 minutes



SUMMARY

Lessons and Next Steps

- A clean manual run is the benchmark and test set
- GPS shifts up to 15 m between days: (Will be ruled out in future)
- LiDAR + gyro carry localisation; walls re-anchor heading
- Measure the vehicle first: the controller must match what the chassis can do
- Planned to have replay logs in a simulator before every flash

8 · Summary

GOAL: Summarise what we learned, what we will change, and open the discussion.

WHAT TO SAY

To close, here are the five lessons from these two field tests.

First, a clean manual run is our benchmark and our test set. log_009 told us where the walls are, where the ramps are and what the sensors report on the correct path. It gave us the route, and it is the yardstick for every later run. We will record more such runs, with Mark presses at every landmark, to check how repeatable the data are.

Second, GPS shifted by up to 15 metres between runs, even with excellent HDOP. A Kalman filter or longer averaging would not help, because this is a bias, not random noise. So GPS will not steer the rover point by point any more. We will use it only for a coarse check: is the rover roughly in the right part of the route, and as a fallback to re-orient when the rover is completely lost.

Third, LiDAR and the gyro carry the localisation. Building walls are measured by the LiDAR to a few centimetres, and they do not move between runs. The gyro gives a smooth heading but drifts slowly; every time the rover drives along a wall of known direction, the wall angle corrects that drift. That is the idea of re-anchoring the heading on walls.

Fourth, measure the vehicle first. log_010 showed that a controller designed for an ideal vehicle does not work on this skid-steered chassis: small steering differences do nothing, and some commands do not move the rover at all. The controller has to be built around what the chassis can physically do, and we measure that from our logs.

Fifth, test in simulation before every flash. We are building Python simulators that replay the logged data and include the measured rover behaviour, so we can find logic errors on the laptop instead of on the campus.

As next steps: adapt the steering to the measured chassis behaviour, replace the rover, which was damaged in a fall during a bench test, record new teaching runs, and run the next field test.

Thank you. We are happy to take your questions.

KEY FACTS TO REMEMBER

Benchmark run · GPS bias up to 15 m → coarse check only · LiDAR + gyro, re-anchored on walls · measure the chassis first · simulate before flashing.

IF YOU ARE ASKED

"When is the next test?" As soon as the replacement rover is available.

"Have you tested anything since log_010?" We have further test data we are still evaluating.

AFTER THE TALK

Collect questions you could not answer and send them to the whole team.